DBMS

Assignment 2

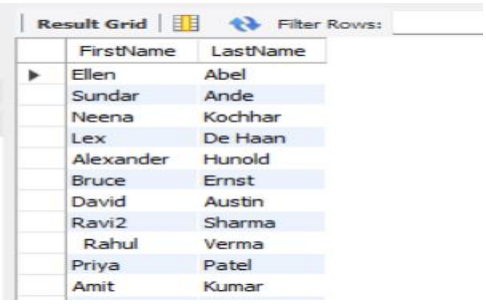
Questions:-

(DATABASE : hr , with table employees having (employee_id, first_name, last_name, salary, department_id, job_id)

Q1. Display first name and last name using aliases

Ans:
```
SELECT first_name AS First Name,
    last_name AS Last Name
FROM employees;
```

Output:



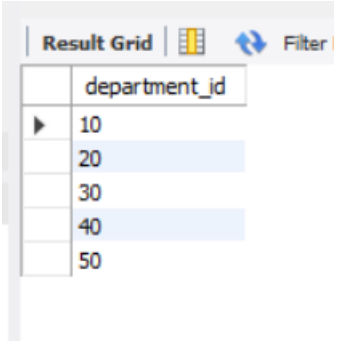
Result Grid | Filter Rows:

FirstName	LastName
Ellen	Abel
Sundar	Ande
Neena	Kochhar
Lex	De Haan
Alexander	Hunold
Bruce	Ernst
David	Austin
Ravi2	Sharma
Rahul	Verma
Priya	Patel
Amit	Kumar

Q2. Get unique department IDs

Ans:
```
SELECT DISTINCT department_id
FROM employees;
```

Output:



Result Grid | Filter

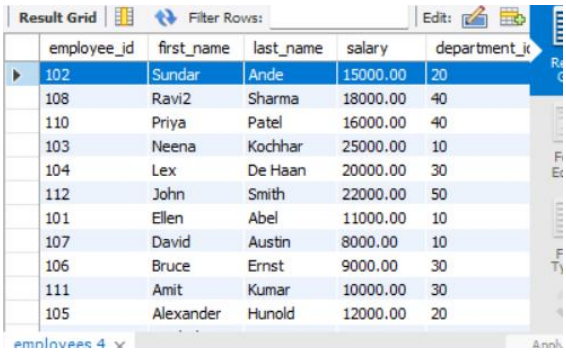
department_id
10
20
30
40
50

Q3. Display all employee details in descending order of first name

Ans:

```
SELECT *
FROM employees
ORDER BY first_name DESC;
```

Output:



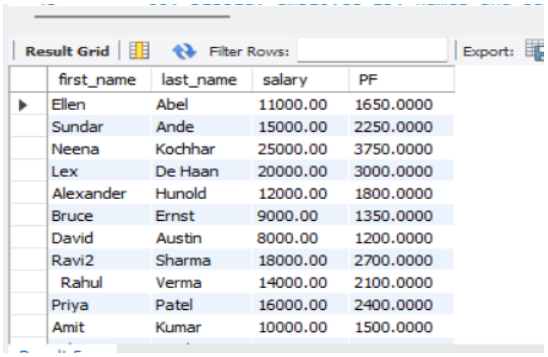
employee_id	first_name	last_name	salary	department_id
102	Sundar	Ande	15000.00	20
108	Ravi2	Sharma	18000.00	40
110	Priya	Patel	16000.00	40
103	Neena	Kochhar	25000.00	10
104	Lex	De Haan	20000.00	30
112	John	Smith	22000.00	50
101	Ellen	Abel	11000.00	10
107	David	Austin	8000.00	10
106	Bruce	Ernst	9000.00	30
111	Amit	Kumar	10000.00	30
105	Alexander	Hunold	12000.00	20

Q4. Display names, salary and PF (15% of salary)

Ans:

```
SELECT first_name, last_name, salary,
    salary * 0.15 AS PF
FROM employees;
```

Output:



first_name	last_name	salary	PF
Ellen	Abel	11000.00	1650.0000
Sundar	Ande	15000.00	2250.0000
Neena	Kochhar	25000.00	3750.0000
Lex	De Haan	20000.00	3000.0000
Alexander	Hunold	12000.00	1800.0000
Bruce	Ernst	9000.00	1350.0000
David	Austin	8000.00	1200.0000
Ravi2	Sharma	18000.00	2700.0000
Rahul	Verma	14000.00	2100.0000
Priya	Patel	16000.00	2400.0000
Amit	Kumar	10000.00	1500.0000

Q5. Display employee ID, names and salary in ascending order

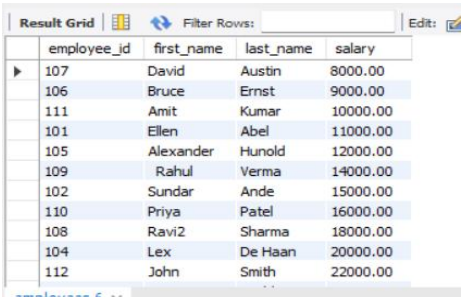
Ans:

```
SELECT employee_id, first_name, last_name, salary
```

FROM employees

ORDER BY salary ASC;

Output:



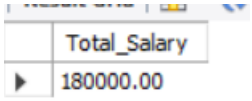
employee_id	first_name	last_name	salary
107	David	Austin	8000.00
106	Bruce	Ernst	9000.00
111	Amit	Kumar	10000.00
101	Ellen	Abel	11000.00
105	Alexander	Hunold	12000.00
109	Rahul	Verma	14000.00
102	Sundar	Ande	15000.00
110	Priya	Patel	16000.00
108	Ravi2	Sharma	18000.00
104	Lex	De Haan	20000.00
112	John	Smith	22000.00

Q6. Calculate total salary

SELECT SUM(salary) AS Total_Salary

FROM employees;

Output:



Total_Salary
180000.00

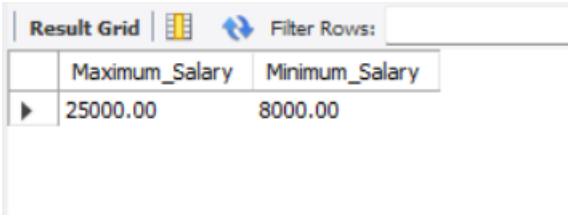
Q7. Find maximum and minimum salary

Ans: SELECT MAX(salary) AS Maximum_Salary,

MIN(salary) AS Minimum_Salary

FROM employees;

Output:



Maximum_Salary	Minimum_Salary
25000.00	8000.00

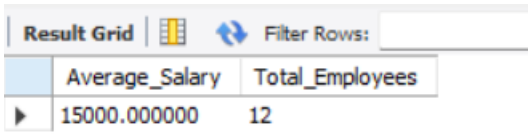
Q8. Find average salary and number of employees

ANS: SELECT AVG(salary) AS Average_Salary,

COUNT(*) AS Total_Employees

FROM employees;

Output:



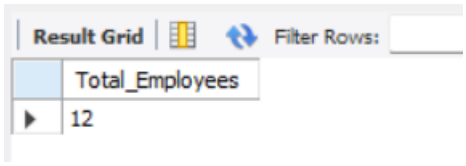
Average_Salary	Total_Employees
15000.000000	12

Q9. Count total employees

Ans: SELECT COUNT(*) AS Total_Employees

FROM employees;

Output:



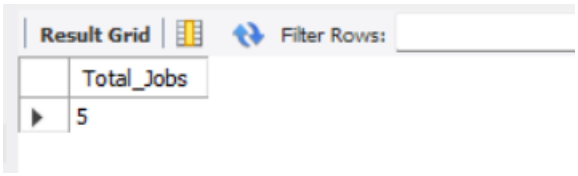
Total_Employees
12

Q10. Count unique jobs

Ans: SELECT COUNT(DISTINCT job_id) AS Total_Jobs

FROM employees;

Output:



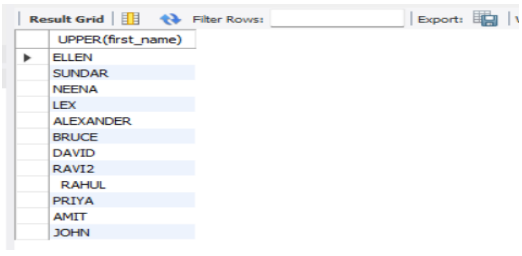
Total_Jobs
5

Q11. Display first names in uppercase

Ans: SELECT UPPER(first_name)

FROM employees;

Output:



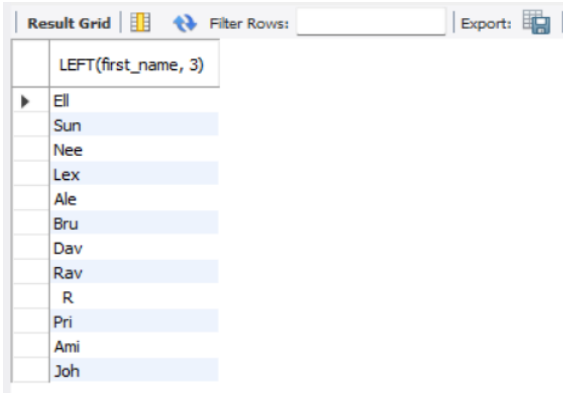
UPPER(first_name)
ELLEN
SUNDAR
NEENA
LEX
ALEXANDER
BRUCE
DAVID
RAVI2
RAHUL
PRIYA
AMIT
JOHN

Q12. Display first 3 characters of first name

Ans: SELECT LEFT(first_name, 3)

FROM employees;

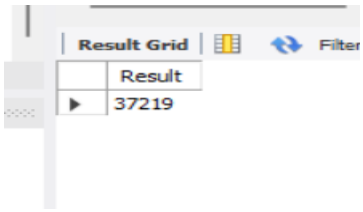
Output:



Q13. Calculate expression

Ans: SELECT 171 * 214 + 625 AS Result;

Output:

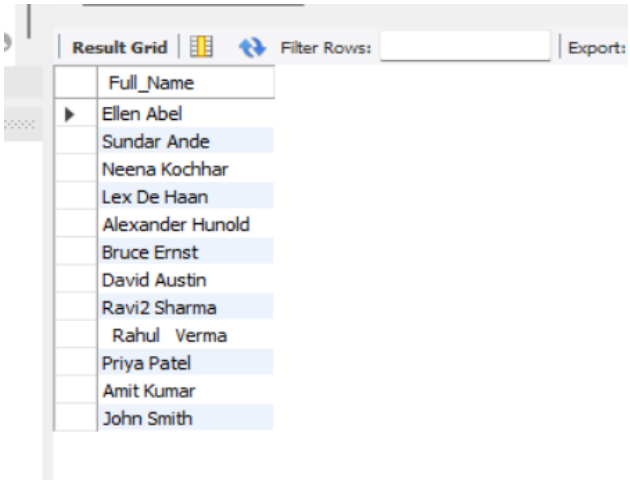


Q14. Display full names of employees

Ans: SELECT CONCAT(first_name, ' ', last_name) AS Full_Name

FROM employees;

Output:

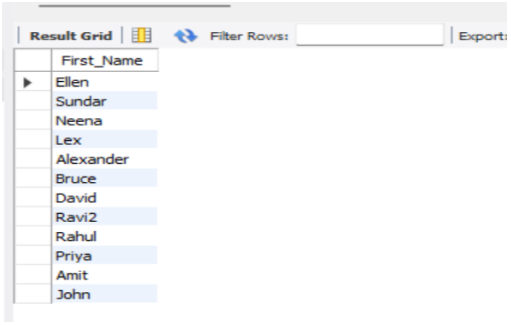


Q15. Remove white spaces from both sides

Ans: SELECT TRIM(first_name) AS First_Name

FROM employees;

Output:



First_Name
Ellen
Sundar
Neena
Lex
Alexander
Bruce
David
Ravi2
Rahul
Priya
Amit
John

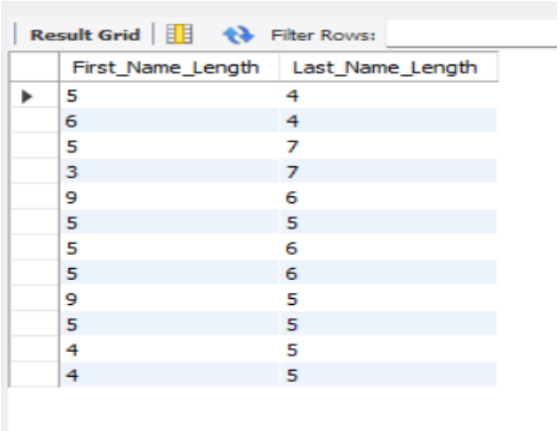
Q16. Find length of first name and last name

Ans: SELECT LENGTH(first_name) AS First_Name_Length,

LENGTH(last_name) AS Last_Name_Length

FROM employees;

Output:



First_Name_Length	Last_Name_Length
5	4
6	4
5	7
3	7
9	6
5	5
5	6
5	6
9	5
5	5
4	5
4	5

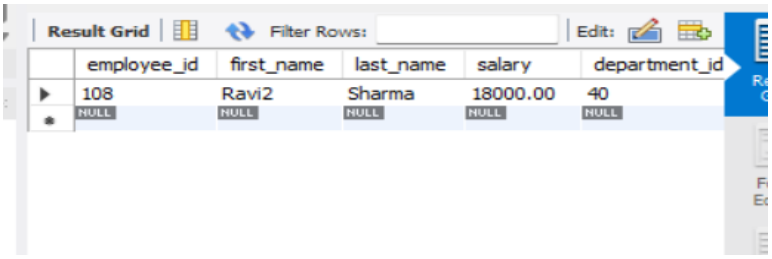
Q17. Find first names containing numbers

Ans: SELECT *

FROM employees

WHERE first_name REGEXP '[0-9]';

Output:



employee_id	first_name	last_name	salary	department_id
108	Ravi2	Sharma	18000.00	40
NULL	NULL	NULL	NULL	NULL

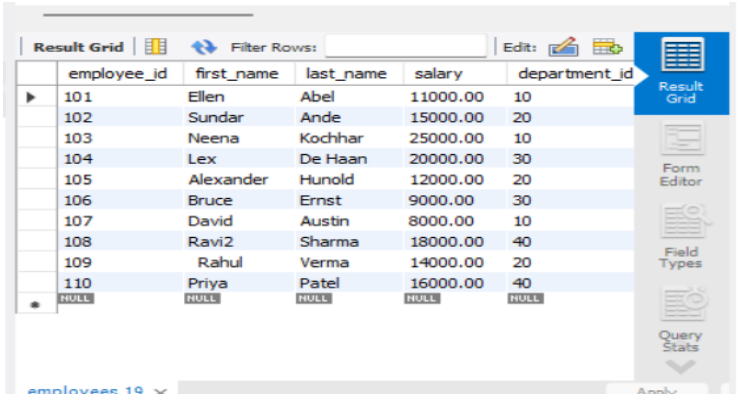
Q18. Display first 10 records

Ans: SELECT *

FROM employees

LIMIT 10;

Output:



employee_id	first_name	last_name	salary	department_id
101	Ellen	Abel	11000.00	10
102	Sundar	Ande	15000.00	20
103	Neena	Kochhar	25000.00	10
104	Lex	De Haan	20000.00	30
105	Alexander	Hunold	12000.00	20
106	Bruce	Ernst	9000.00	30
107	David	Austin	8000.00	10
108	Ravi2	Sharma	18000.00	40
109	Rahul	Verma	14000.00	20
110	Priya	Patel	16000.00	40
NULL	NULL	NULL	NULL	NULL

Q19. Calculate monthly salary rounded to 2 decimal places

Ans: SELECT employee_id, first_name, last_name,

ROUND(salary / 12, 2) AS Monthly_Salary

FROM employees;

Output:


Result Grid | Filter Rows: | Export:

employee_id	first_name	last_name	Monthly_Salary
101	Ellen	Abel	916.67
102	Sundar	Ande	1250.00
103	Neena	Kochhar	2083.33
104	Lex	De Haan	1666.67
105	Alexander	Hunold	1000.00
106	Bruce	Ernst	750.00
107	David	Austin	666.67
108	Ravi2	Sharma	1500.00
109	Rahul	Verma	1166.67
110	Priya	Patel	1333.33
111	Amit	Kumar	833.33
112	John	Smith	1833.33

Result 20